PROJECT REPORT

Waste Management Data Warehouse & ETL Pipeline

Designing a scalable data warehouse and ETL pipeline for transforming raw data into actionable insights.

By
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Abstract

This report documents the design and implementation of a data warehousing project titled **"Data Warehouse Development & ETL Processing: Q100 Waste Data Flow."** The project is based on the Q100 Waste data published by the Department for Environment, Food and Rural Affairs (DEFRA) and focuses on creating a prototype data warehouse for Local Authority Waste Management. These Q100 Waste datasets, covering two distinct fiscal years, are the foundation for building a unified data warehouse to enhance analytical capabilities and reporting for waste management. Utilizing Kimball's Dimensional Modelling Technique, a star schema is designed to support analytical queries and reporting requirements. Utilizing Kimball's Dimensional Modelling Technique, a star schema is designed to align with best practices in data warehousing, ensuring the efficient organization of data into fact and dimension tables.The project includes tasks such as ETL processing using SQL Server Integration Services (SSIS), data cleaning, transformations, and dashboard development in Power BI. The primary objective was to process and transform disparate datasets from two source files into a unified data warehouse capable of supporting analytical and decision-making processes for waste management authorities.The project's significance lies in addressing data discrepancies, performing transformations, and providing actionable insights via visual analytics. Challenges encountered during the project, such as column mismatches and null values, were meticulously handled using data cleaning and transformation techniques. The report concludes with a reflection on the project's outcomes, its alignment with data management best practices, and opportunities for future enhancements.

I. Background

Data warehousing is pivotal in modern data management, especially when dealing with large and diverse datasets. By transforming raw data into an integrated and structured form, data warehouses enable efficient querying and analysis, driving informed decision-making. In this project, a prototype data warehouse was designed using Kimball's Dimensional Modelling Technique, ensuring an optimized schema for analytical purposes. The **"Data Warehouse Development & ETL Processing: Q100 Waste Data Flow"** project stems from the increasing need to manage and analyze complex datasets in waste management effectively. The Q100 Waste data, published by the Department for Environment, Food and Rural Affairs (DEFRA), serves as a comprehensive source of information on waste management statistics for Local Authorities. The data, spanning two fiscal years (2015-2016 and 2016-2017), contains valuable insights but also presents significant challenges due to discrepancies in its structure and missing data thus requiring significant preprocessing to harmonize their structure.

The background of this project also highlights the technical tools and setup required for its execution. Key software, including SQL Server, SSMS, SSIS, and Visual Studio, were utilized to build and manage the ETL pipeline and data warehouse infrastructure. The integration of SSIS allowed for efficient data transformation, cleaning, and loading processes, laying the groundwork for advanced data warehousing operations. This project demonstrates the crucial role of data warehouses in integrating and transforming datasets into a cohesive structure for analytical purposes. By addressing the challenges posed by the Q100 Waste data, the project provides a valuable prototype for managing waste data efficiently, setting the stage for further advancements in public sector data management and decision-making.

II. Introduction

Data management has become a cornerstone of modern information systems, providing organizations with the ability to manage, analyze, and derive insights from ever-growing datasets. This project, titled **"Data Warehouse Development & ETL Processing: Q100 Waste Data Flow,"** focuses on designing and implementing a scalable data warehouse solution to support analytical reporting of the concepts, tools, and techniques central to data management roles. The primary objective was to design a prototype data warehouse using the Q100 Waste datasets published by the Department for Environment, Food and Rural Affairs (DEFRA) for Local Authority Waste Management across two fiscal years: 2015-2016 and 2016-2017.

Through this project, key learning outcomes were addressed, including evaluating and implementing best practices in data management, assessing emerging tools and technologies, and fostering a self-directed approach to learning. The project required analyzing real-world datasets to identify inconsistencies, such as mismatched columns and null values, and applying dimensional modeling techniques, particularly Kimball's star schema design, to transform raw data into an analytical format. Tools such as SQL Server Integration Services (SSIS), SQL Server Management Studio (SSMS), and Power BI were leveraged for data integration, transformation, querying, and visualization.

The project's significance lies not only in creating a functional data warehouse but also in demonstrating the critical role of ETL processes, schema design, and dashboard creation in meeting organizational needs for data-driven decision-making. It provided the opportunity to evaluate current data management software and processes, enhance technical proficiency, and apply self-directed learning to solve complex problems. By combining theoretical knowledge with practical skills, this project underscores the importance of structured approaches to data management in addressing modern challenges.

III. Project Objectives:

The main objective of this project is to design and implement a robust data warehouse prototype using the Kimball Dimensional Modeling Technique. The project aims to leverage MS SQL Server, SSIS, and business intelligence tools like Tableau or Power BI to build an efficient system for data analysis and reporting. Specifically, the project seeks to achieve the following objectives:

1. **Design a Data Model Using Kimball's Dimensional Modeling**:
   - Use Kimball's Dimensional Modelling Technique to design a schema diagram (star/snowflake) data model for a prototype data warehouse using the Q100 Waste data for: WasteDataFlow - Local Authority waste management Published by Department for Environment, Food and Rural Affairs.
     http://data.defra.gov.uk/Waste/Q100_data_England_2015_2016.csv
     http://data.defra.gov.uk/Waste/Q100_data_England_2016_2017.csv

2. **Implement the ETL Process Using SSIS**:
   - Extract data from various sources and transform it to match the data warehouse schema.
   - There are column mismatches between the two source files, resolve this so the two files are union-compatible and merge their content, as well as fix any other issues that may come across in the data.
   - Change the ONSCode for" Kent County Council" to student ID.
   - Load the transformed data into the data warehouse.

3. **Develop SQL Queries for Testing and Critical Analysis**:
   - Write SQL queries that involve gathering and analyzing data, ranking items based on certain criteria, grouping data by time-periods, and creating stored procedure to handle different inputs and generate detailed reports.

4. **Create Interactive Dashboards Using Tableau or Power BI**:
   - Use either Tableau or Power BI to design an interactive dashboard combining the query tasks in a single view as visualized elements, that can be used to tell an effective story about the business.

IV. System Design:

A. Install Required Software and Tools

The project is implemented using SQL Server for the data warehouse, SSIS for the ETL process, and Power BI for business intelligence reporting. To successfully execute the project, the following software and tools were installed and configured:

1. **SQL Server Developer Edition**: Installed from the official Microsoft website (SQL Server Downloads) to create and manage the database.

2. **SQL Server Management Studio (SSMS)**: Installed from SSMS Full Setup for managing and querying the SQL Server database.

3. **SQL Server Integration Services (SSIS)**: Configured through the SQL Server Configuration Manager and Visual Studio extensions to handle ETL operations.
4. **Visual Studio**: Downloaded from Visual Studio Downloads and extended with SSIS tools for designing ETL workflows.
5. **Power BI**: Installed to create interactive dashboards and visualize the results derived from the data warehouse. Downloaded from: Download Power BI.

B. Data Exploration

The project began with analyzing two datasets from fiscal years 2015-2016 and 2016-2017. The datasets contained inconsistencies and the primary challenge was the mismatch of columns between the two datasets.

The integration of datasets from fiscal years 2015-2016 and 2016-2017 revealed several inconsistencies that required resolution before proceeding to the star schema design. The primary challenge was the mismatch of columns between the two datasets.

1. Identifying Mismatched Columns

During the exploration phase, the following columns were identified as mismatched or inconsistent between the datasets:

- **FacilityCode**
- **ONSCode**
- **QuarterlyComments**
- **TonnesFromCommercialSources**
- **TonnesFromIndustrialSources**
- **Usage**

2. Resolving Mismatched Columns

To make the datasets compatible for union, the following actions were taken:

- **Dropped Irrelevant Columns**:
  Columns with no data (**QuarterlyComments**, **TonnesFromCommercialSources**, **TonnesFromIndustrialSources**, and **Usage**) were dropped from both files as these columns were deemed irrelevant as they contained no meaningful data in both files.
- **Added Missing Columns with Null Values**:
  For columns that contained data but were missing in one file, null placeholders were added to ensure consistency:
  - **FacilityCode** found in the 2016-2017 dataset but missing in 2015-2016 so it was added to the 2015-2016 dataset.
  - **ONSCode** found in the 2015-2016 dataset but missing in 2016-2017 so it was added to the 2016-2017 dataset.

C. Data Flow in SSIS

An SSIS package was created in Visual Studio to integrate and standardize the source data:

- **Data Flow Creation**: A Data Flow container was added to the package to manage the integration of both datasets.
- **Union Compatibility**: Data transformations were applied to align the structure and schema of the datasets, making them union-compatible using a **Union All Transformation**, the modified datasets were combined into a single data source file, which served as the input for the subsequent data modeling phase.

This meticulous approach ensured data consistency and compatibility, setting the foundation for the star schema design.

D. Identifying Fact and Dimension Tables

After obtaining the single data source file, the attributes were analyzed to identify fact and dimension tables:

- **Fact Table**: A Fact Table was designed to store quantitative data (e.g., tonnage-related metrics).
- **Dimension Tables**: Dimension Tables were created to describe the categorical aspects of the data, such as authority, period, material, facility, waste stream, and output process type.

E. Star Schema Design

The star schema was implemented in SQL Server Management Studio (SSMS) using the following structure:

Fact Table: Fact_WasteManagement

The fact table contains the quantitative data, such as TotalTonnes, TonnesByMaterial, TonnesFromHHSources, TonnesFromNonHHSources, TonnesFromWfHSources, TonnesFromWnfHSources, WasteProcessorId, WasteProcessorOutputId, SenderWasteProcessorOutputId, MonthlyComments. Also, it includes foreign keys that reference the dimension tables, ensuring that the relationships between the data elements are preserved.

Dimension Tables

The dimension tables describe the categorical attributes of the data:

- **Dim_Authority**: Contains information about authorities, including AuthorityID, AuthorityName, and ONSCode.

- **Dim_Period**: Contains PeriodId and Period and derived time-related attributes, such as Year, Quarter, and SixMonthPeriod.
- **Dim_Material**: Describes materials using attributes like Materiald, MaterialName and MaterialGroup.
- **Dim_Facility**: Captures facility details, including FacilityID, FacilityTypeId, FacilityType, NationalFacilityId, FacilityName, FacilityAddress, FacilityPostCode,FacilityLicence
- **Dim_WasteStream**: Represents different types of waste streams with attributes like WasteStreamId, WasteStreamTypeId, WasteStreamType.
- **Dim_OutputProcessType**: Describes output processing types by OutputProcessTypeId and OutputProcessType columns.

Star Schema Implementation in SQL

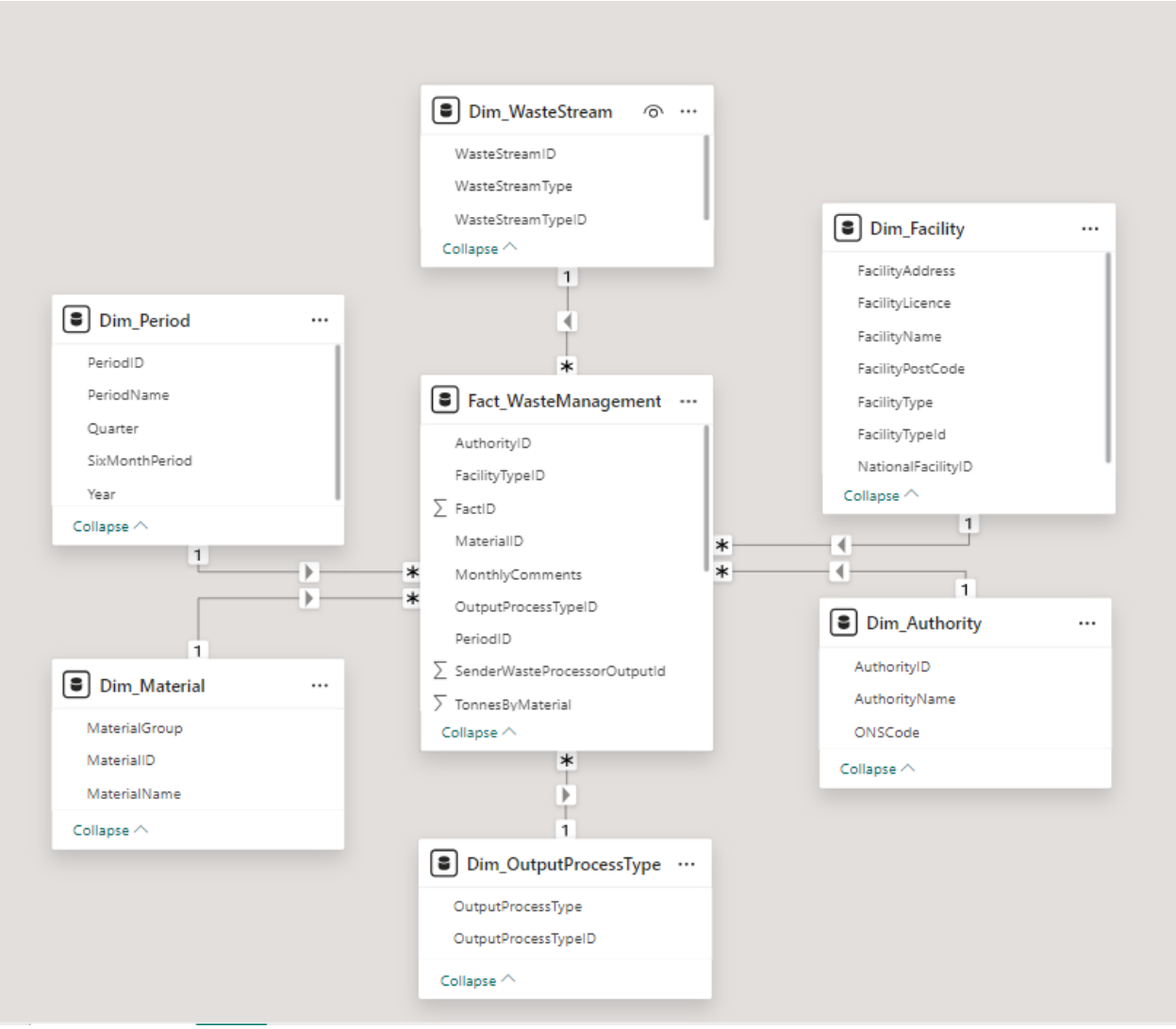
The following SQL commands were used to implement the star schema in SSMS:

```sql
USE Star_Schema;
GO
-- Dim_Authority
CREATE TABLE Dim_Authority(
    AuthorityID INT PRIMARY KEY,
    AuthorityName NVARCHAR(100),
    ONSCode NVARCHAR(50)
);

-- Dim_Period
CREATE TABLE Dim_Period (
    PeriodID INT PRIMARY KEY,
    PeriodName NVARCHAR(100),
    Year INT,
    Quarter NVARCHAR(2),
    SixMonthPeriod NVARCHAR(10)
);
-- Dim_Material
CREATE TABLE Dim_Material (
    MaterialID INT PRIMARY KEY,
    MaterialName NVARCHAR(100),
    MaterialGroup NVARCHAR(100)
);
-- Dim_Facility
CREATE TABLE Dim_Facility (
    FacilityTypeId INT PRIMARY KEY,
    FacilityType NVARCHAR(225),
    NationalFacilityID INT,
    FacilityName NVARCHAR(225),
    FacilityAddress NVARCHAR(225),
    FacilityPostCode NVARCHAR(225),
    FacilityLicence NVARCHAR(225)
);
-- Dim_WasteStream
CREATE TABLE Dim_WasteStream (
    WasteStreamID INT PRIMARY KEY,
    WasteStreamTypeID INT,
    WasteStreamType NVARCHAR(100)
);
-- Dim_OutputProcessType
CREATE TABLE Dim_OutputProcessType (
    OutputProcessTypeID INT PRIMARY KEY,
    OutputProcessType NVARCHAR(225)
);
```

```sql
--Fact Table
-- Fact_WasteManagement
CREATE TABLE Fact_WasteManagement (
    FactID INT PRIMARY KEY,
    AuthorityID INT,
    PeriodID INT,
    WasteProcessorID INT,
    WasteProcessorOutputId INT,
    SenderWasteProcessorOutputId INT,
    OutputProcessTypeID INT,
    WasteStreamID INT,
    FacilityTypeID INT,
    MaterialID INT,
    TotalTonnes FLOAT,
    TonnesByMaterial FLOAT,
    TonnesFromHHSources FLOAT,
    TonnesFromNonHHSources FLOAT,
    TonnesFromWfHSources FLOAT,
    TonnesFromWnfHSources FLOAT,
    MonthlyComments NVARCHAR(500)
);

-- Adding Foreign Keys
ALTER TABLE Fact_WasteManagement
ADD CONSTRAINT FK_Authority FOREIGN KEY (AuthorityID) REFERENCES Dim_Authority(AuthorityID);

ALTER TABLE Fact_WasteManagement
ADD CONSTRAINT FK_WasteStream FOREIGN KEY (WasteStreamID) REFERENCES Dim_WasteStream(WasteStreamID);

ALTER TABLE Fact_WasteManagement
ADD CONSTRAINT FK_FacilityType FOREIGN KEY (FacilityTypeID) REFERENCES Dim_Facility(FacilityTypeID);

ALTER TABLE Fact_WasteManagement
ADD CONSTRAINT FK_OutputProcessType FOREIGN KEY (OutputProcessTypeID) REFERENCES Dim_OutputProcessType(OutputProcessTypeID);

ALTER TABLE Fact_WasteManagement
ADD CONSTRAINT FK_Material FOREIGN KEY (MaterialID) REFERENCES Dim_Material(MaterialID);

ALTER TABLE Fact_WasteManagement
ADD CONSTRAINT FK_Period FOREIGN KEY (PeriodID) REFERENCES Dim_Period(PeriodID);
```

Joins additional columns to the

This design ensures that the star schema is highly optimized for query performance and analytical processing.



Star Schema Diagram

This structured system design enabled the data warehouse to support efficient data retrieval, reporting, and analytics.

V. Implementation:

The implementation phase consisted of transforming the prepared dataset into a star schema in SQL Server Management Studio (SSMS) using SQL Server Integration Services (SSIS). Each dimension and fact table was populated through an Extract, Transform, Load (ETL) pipeline, addressing challenges in data cleaning, transformation, and relational mapping.

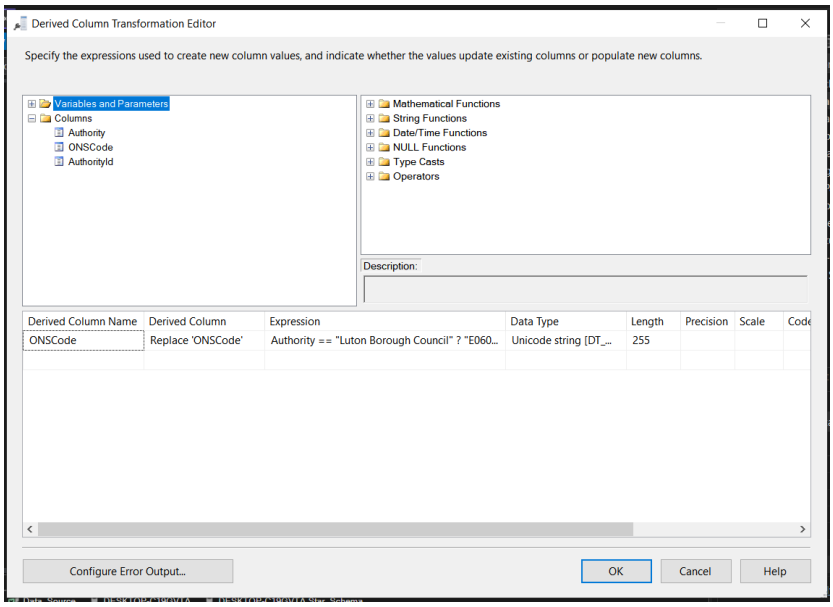
A. ETL Using SSIS

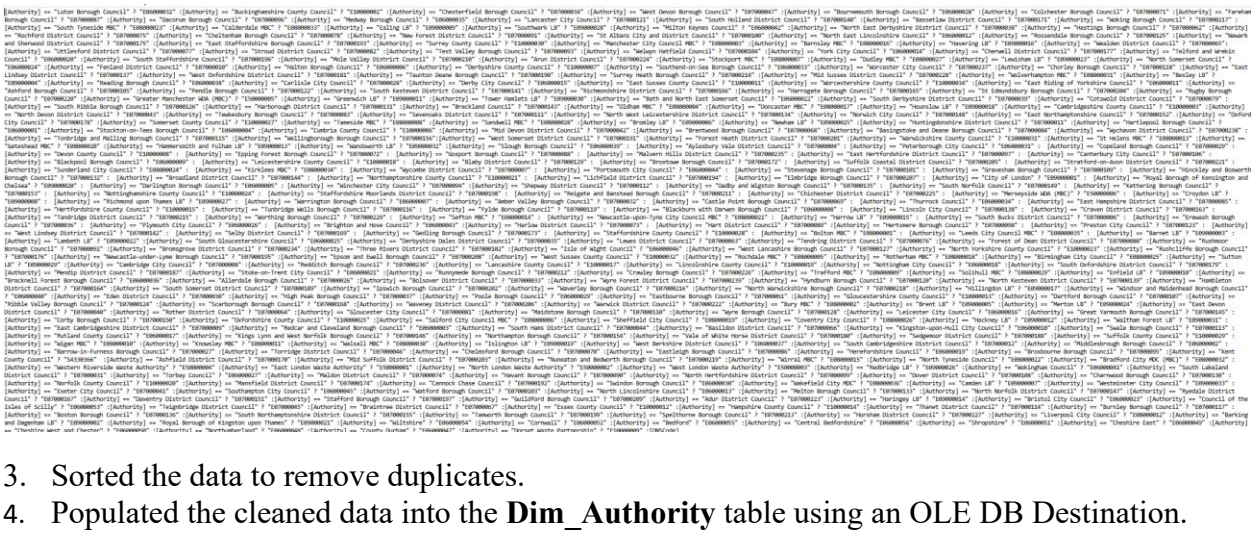
Dimension Tables

Dim_Authority Table:

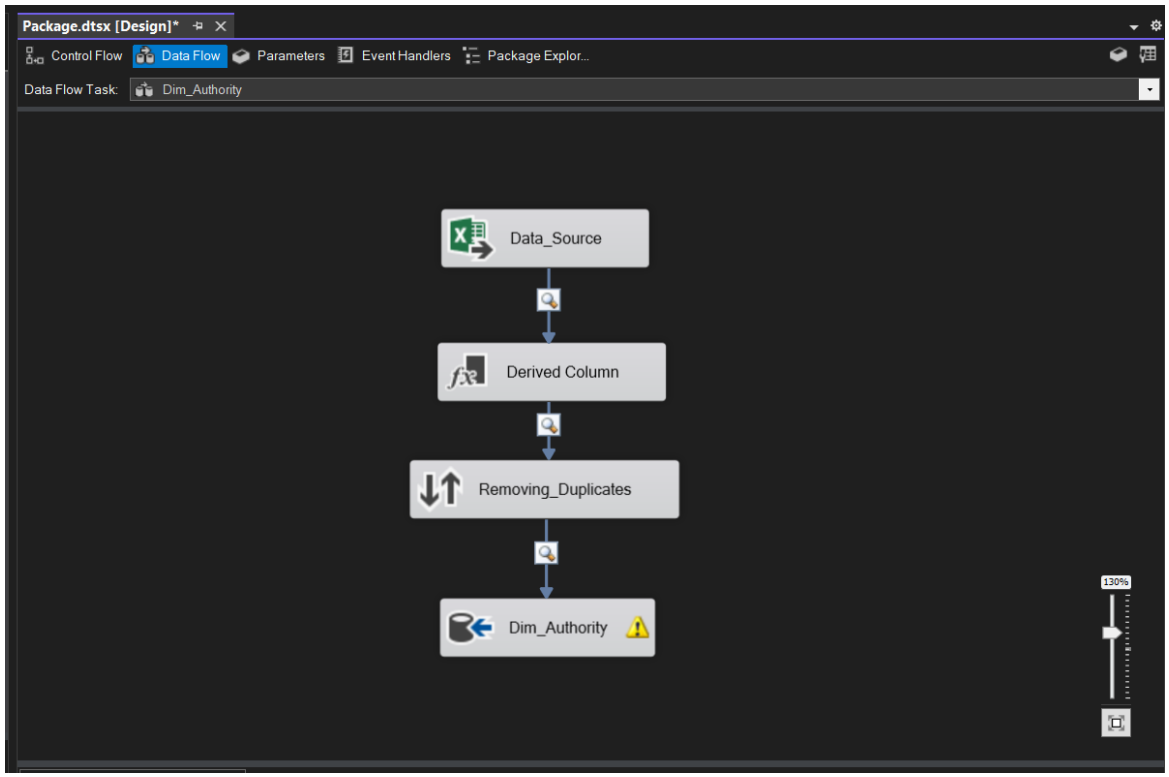
1. Loaded the single data source file using the connection manager to the source file.

2. Used a **Derived Column** transformation to handle null values in the ONSCode column:
   - Replaced null ONSCode values by comparing them with the Authority column.
   - Resolved the challenge of writing a conditional expression in Visual Studio 2022, which doesn't support IIF in this scenario, with a nested ternary operator.
   - Updated the `ONSCode` for "Kent County Council" to the student ID `S4330366`.



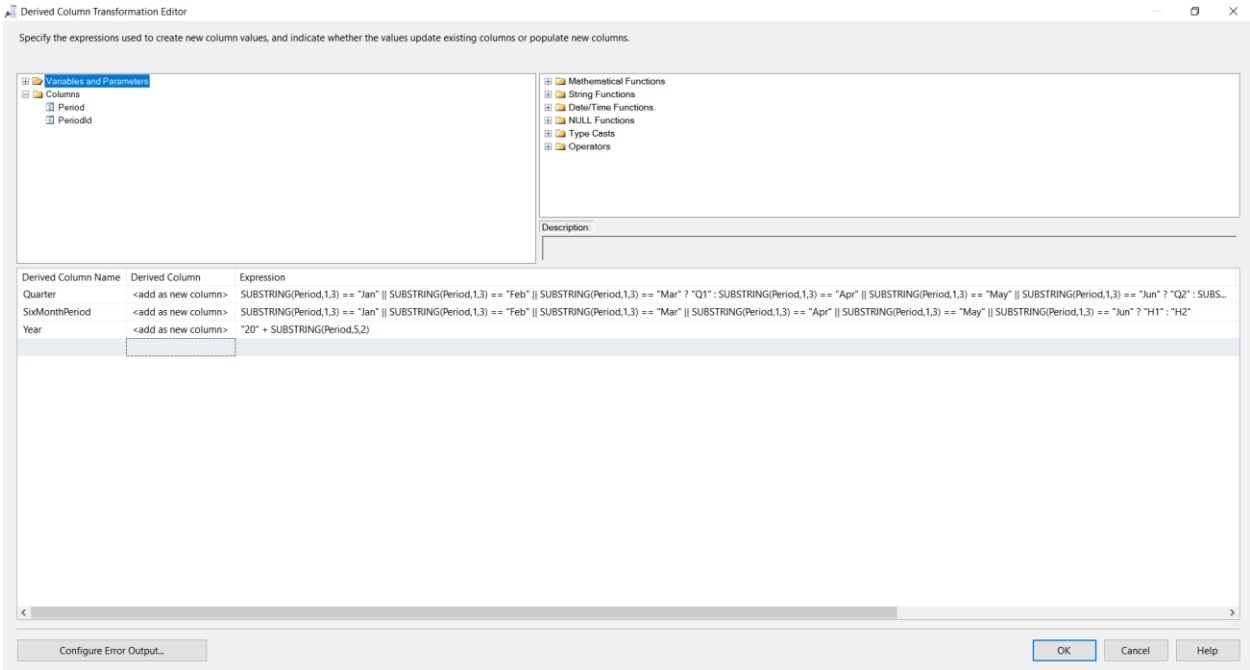


3. Sorted the data to remove duplicates.
4. Populated the cleaned data into the **Dim_Authority** table using an OLE DB Destination.

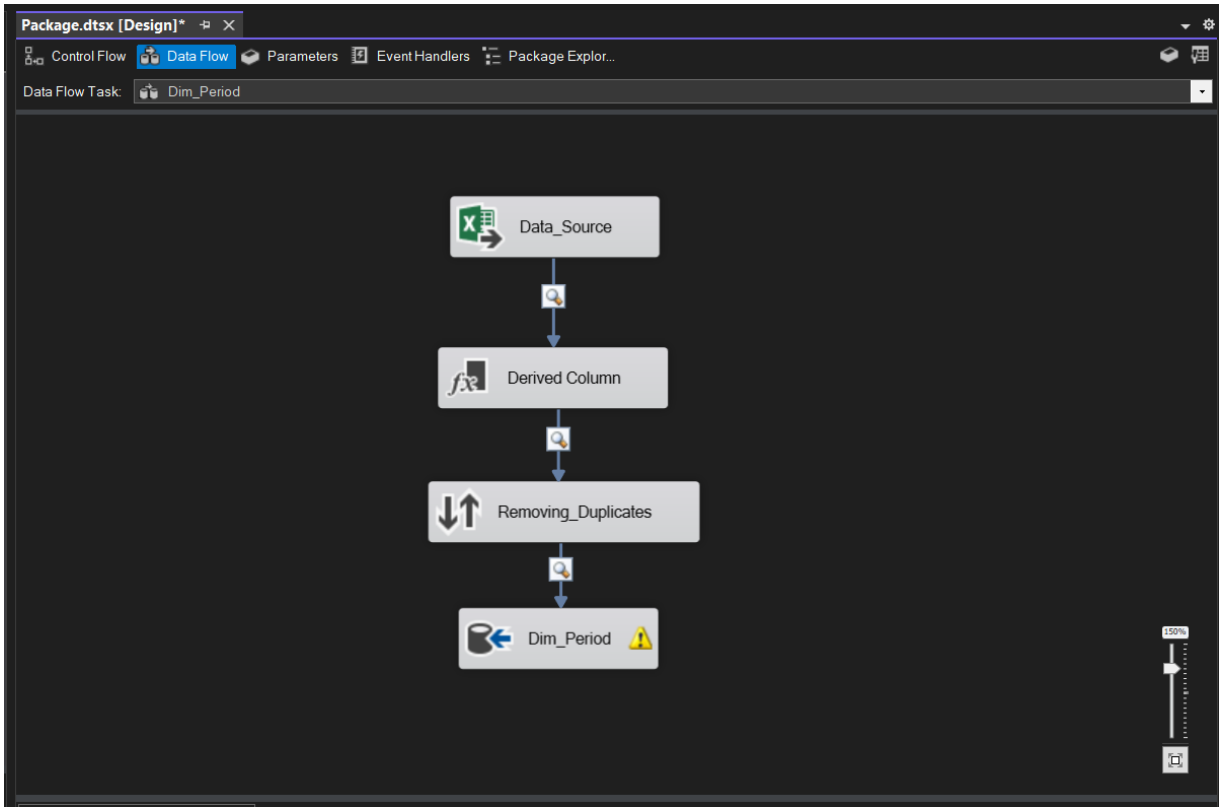


Dim_Period Table:

1. Extracted and transformed Period data into three new columns using Derived Column transformations:



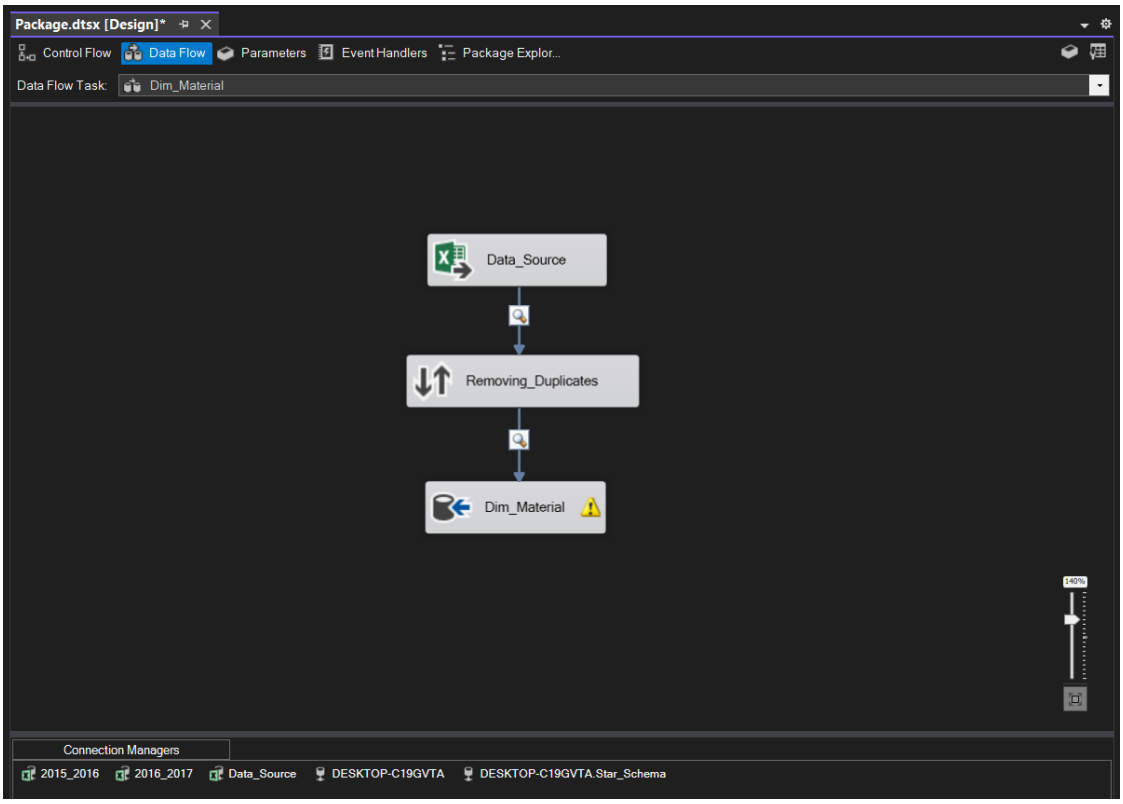
2. Removed duplicates for better efficiency.
3. Loaded the transformed data into the **Dim_Period** table.

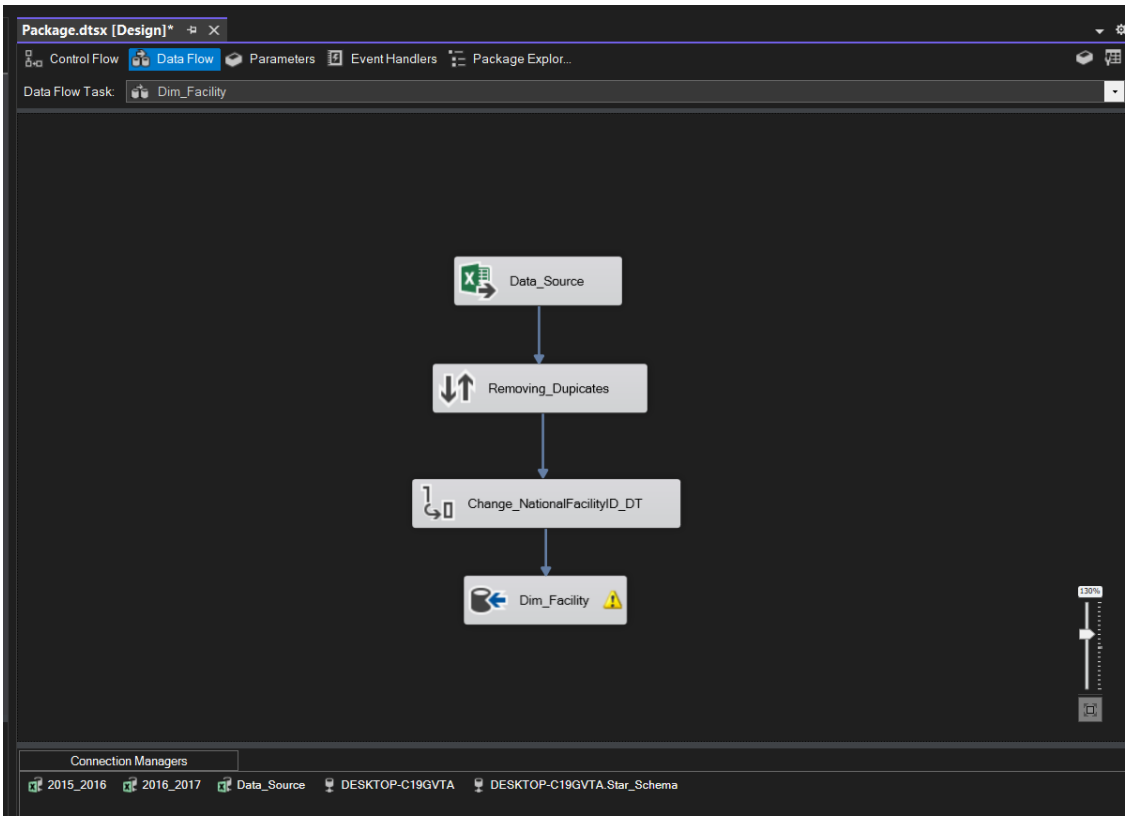


Other Dimension Tables (Dim_Material, Dim_OutputProcessType, Dim_WasteStream, Dim_Facility):

- For each table, the process was as follows:

1. Loaded the data from the source file.
2. Performed **Data Conversions** and removed duplicates using a **Sort** container.
3. Populate the cleaned data into the respective dimension table.


Package.dtsx [Design]*
Control Flow
Data Flow
Parameters
Event Handlers
Package Explor...
Data Flow Task:
Dim_Material
Data_Source
Removing_Duplicates
Dim_Material
140%
Connection Managers
2015_2016
2016_2017
Data_Source
DESKTOP-C19GVTA
DESKTOP-C19GVTA.Star_Schema


Package.dtsx [Design]*
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Package Explor...
Data Flow Task:
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Data_Source
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Change_NationalFacilityID_DT
Dim_Facility
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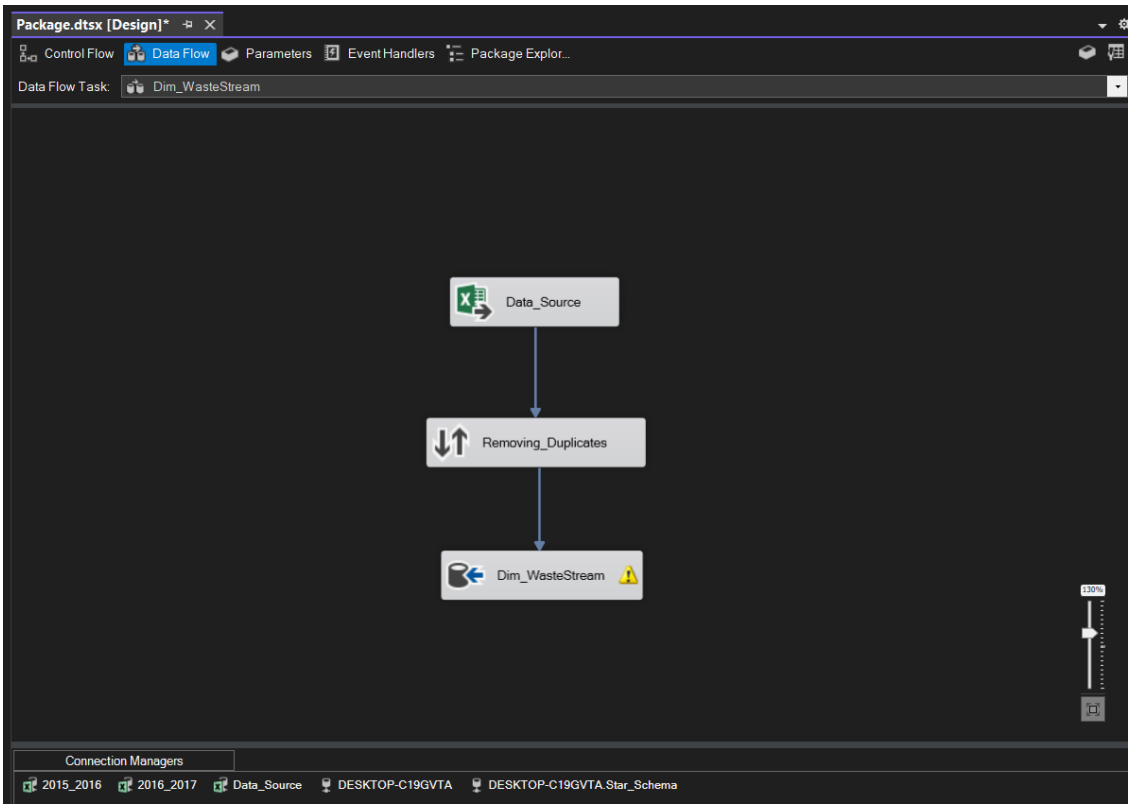
Connection Managers
2015_2016
2016_2017
Data_Source
DESKTOP-C19GVTA
DESKTOP-C19GVTA.Star_Schema


Package.dtsx [Design]*
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Package Explor...
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Dim_WasteStream
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2015_2016
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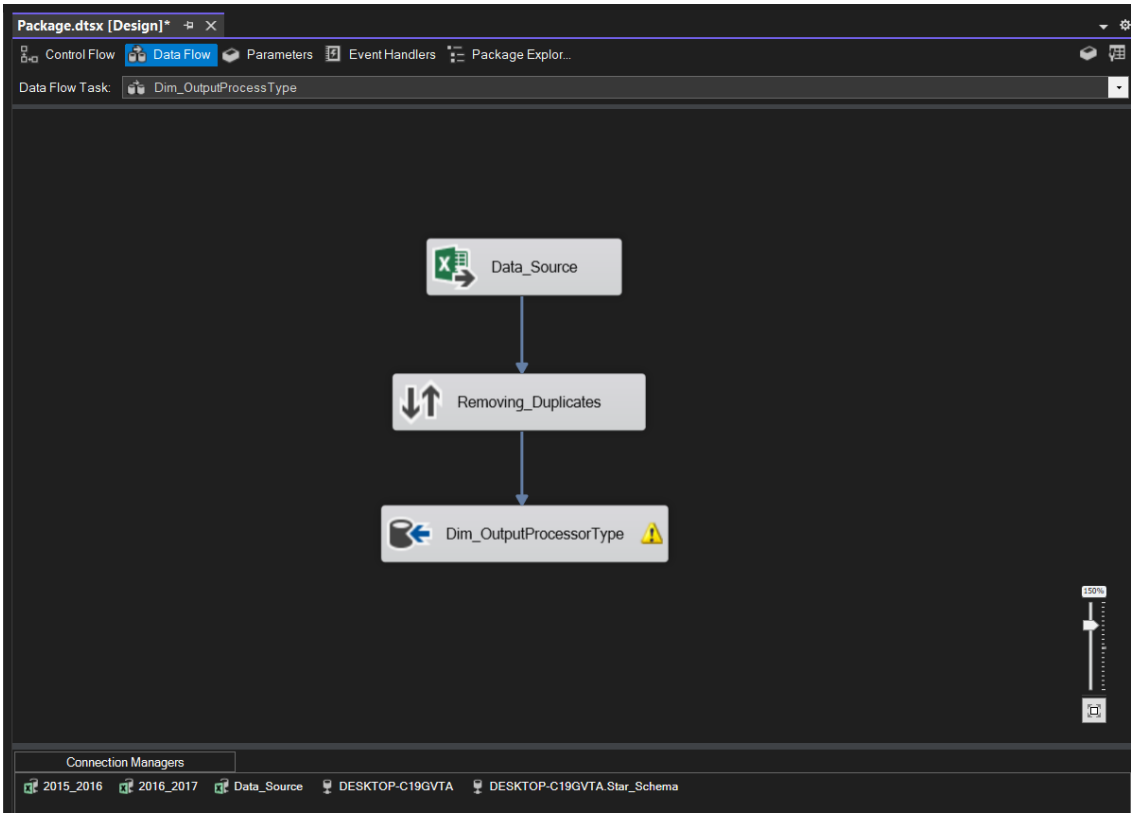
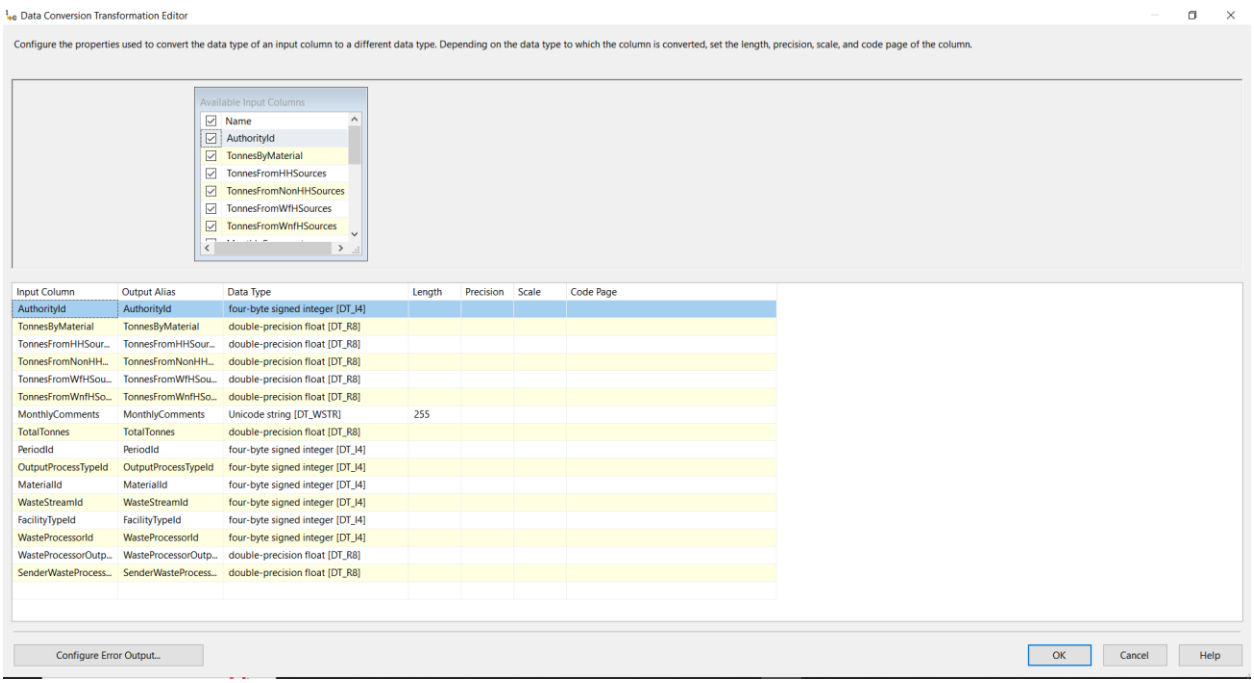

Package.dtsx [Design]*
Control Flow
Data Flow
Parameters
Event Handlers
Package Explor...
Data Flow Task:
Dim_OutputProcessType
Data_Source
Removing_Duplicates
Dim_OutputProcessorType
150%
Connection Managers
2015_2016
2016_2017
Data_Source
DESKTOP-C19GVTA
DESKTOP-C19GVTA.Star_Schema

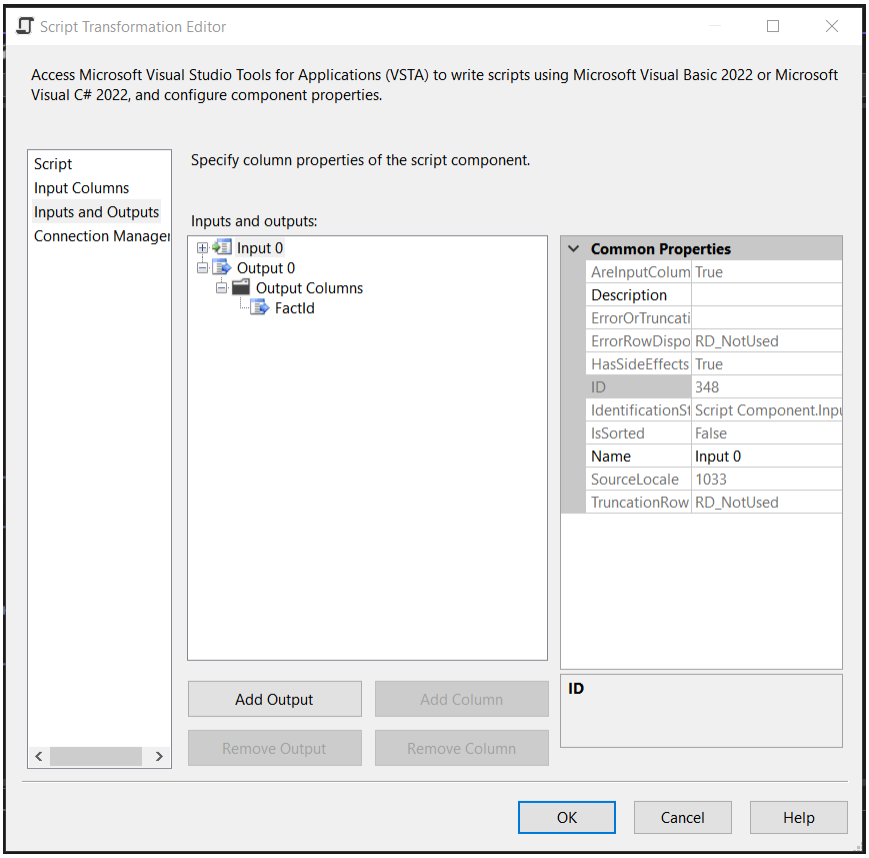
Fact_WasteManagement Table

1. Loaded the single data source using the connection manager.
2. Performed **Data Conversions** for the fact table columns:
   - Used **DT_UI4 (4-byte unsigned integer)** for ID columns (AuthorityId, PeriodId, WasteProcessorId, WasteProcessorOutputId, senderWasteProcessorOutputId, WasteStreamTypeId, FacilityTypeId, OutputProcessTypeId, MaterialId, NationalFacilityId, WasteStreamId) to optimize memory usage.
   - Used **Double Precision Float** for tonnage-related columns (TotalTonnes, TonnesByMaterial, TonnesFromHHSources, TonnesFromNonHHSources ,TonnesFromWfHSources, TonnesFromWnfHSources, WasteProcessorId, WasteProcessorOutputId, SenderWasteProcessorOutputId, MonthlyComments).



Input Column	Output Alias	Data Type	Length	Precision	Scale	Code Page
AuthorityId	AuthorityId	four-byte signed integer [DT_I4]				
TonnesByMaterial	TonnesByMaterial	double-precision float [DT_R8]				
TonnesFromHHSour...	TonnesFromHHSour...	double-precision float [DT_R8]				
TonnesFromNonHH...	TonnesFromNonHH...	double-precision float [DT_R8]				
TonnesFromWfHSou...	TonnesFromWfHSou...	double-precision float [DT_R8]				
TonnesFromWnfHSo...	TonnesFromWnfHSo...	double-precision float [DT_R8]				
MonthlyComments	MonthlyComments	Unicode string [DT_WSTR]	255			
TotalTonnes	TotalTonnes	double-precision float [DT_R8]				
PeriodId	PeriodId	four-byte signed integer [DT_I4]				
OutputProcessTypeId	OutputProcessTypeId	four-byte signed integer [DT_I4]				
MaterialId	MaterialId	four-byte signed integer [DT_I4]				
WasteStreamId	WasteStreamId	four-byte signed integer [DT_I4]				
FacilityTypeId	FacilityTypeId	four-byte signed integer [DT_I4]				
WasteProcessorId	WasteProcessorId	four-byte signed integer [DT_I4]				
WasteProcessorOutp...	WasteProcessorOutp...	double-precision float [DT_R8]				
SenderWasteProcess...	SenderWasteProcess...	double-precision float [DT_R8]				

3. Used a **Script Transformation** to generate a unique Fact_Id for each record.


Script Transformation Editor
Access Microsoft Visual Studio Tools for Applications (VSTA) to write scripts using Microsoft Visual Basic 2022 or Microsoft Visual C# 2022, and configure component properties.
Script
Input Columns
Inputs and Outputs
Connection Manager
Specify column properties of the script component.
Inputs and outputs:
Input 0
Output 0
Output Columns
FactId
Common Properties
AreInputColum True
Description
ErrorOrTruncati
ErrorRowDispo RD_NotUsed
HasSideEffects True
ID 348
IdentificationSt Script Component.Inpu
IsSorted False
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ID
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Add Column
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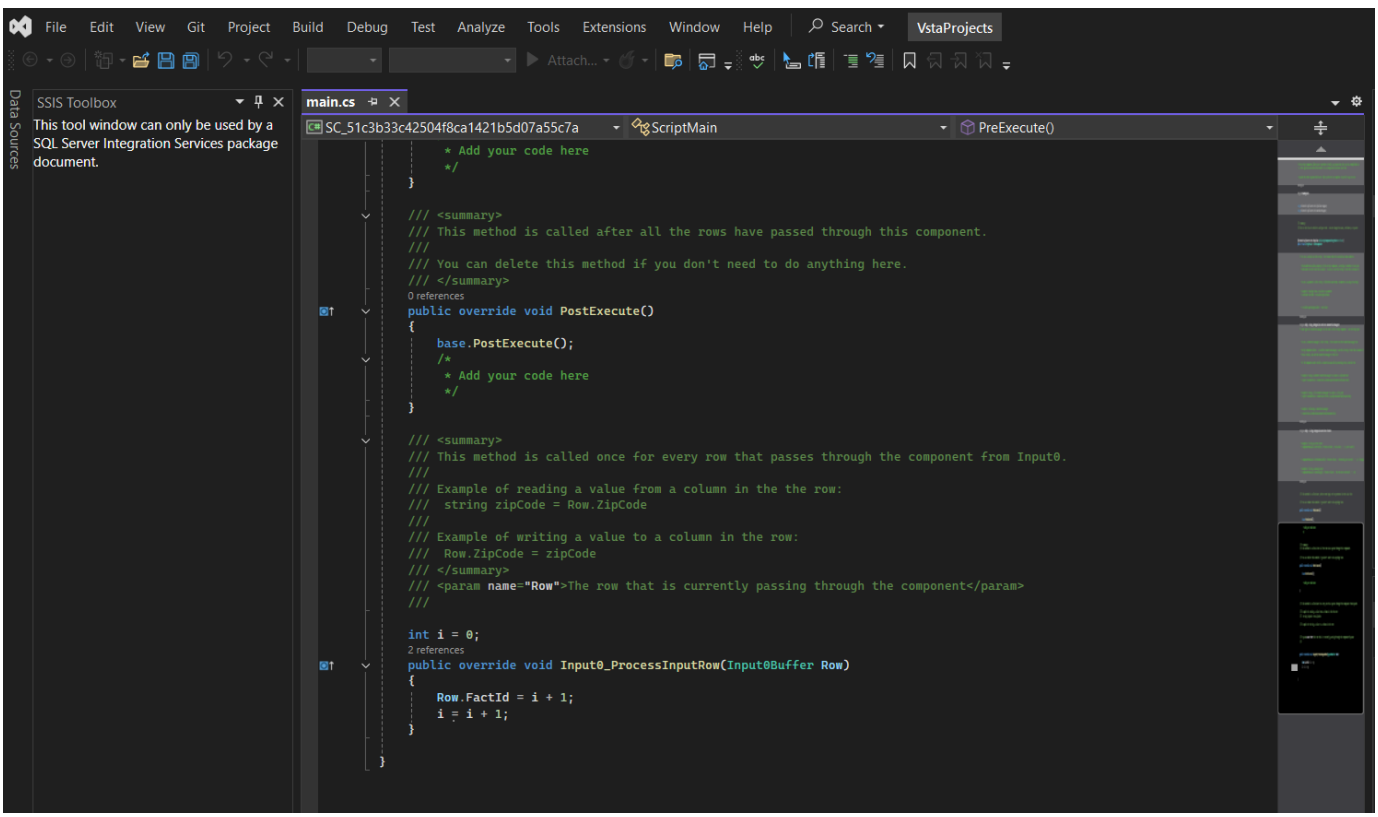
File Edit View Git Project Build Debug Test Analyze Tools Extensions Window Help
Search
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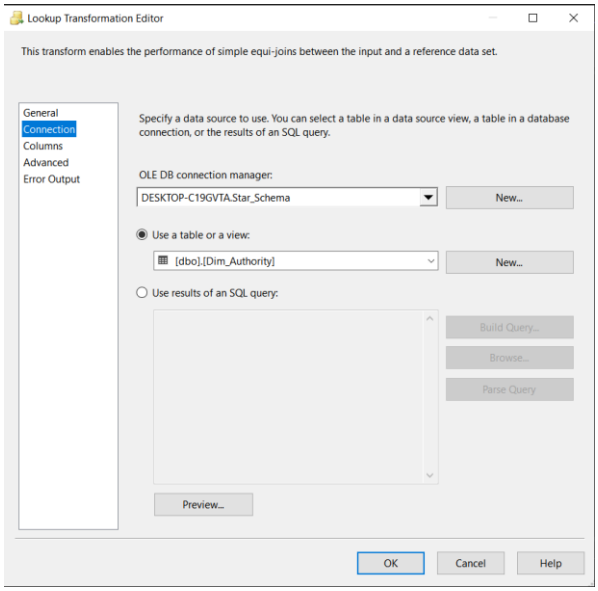
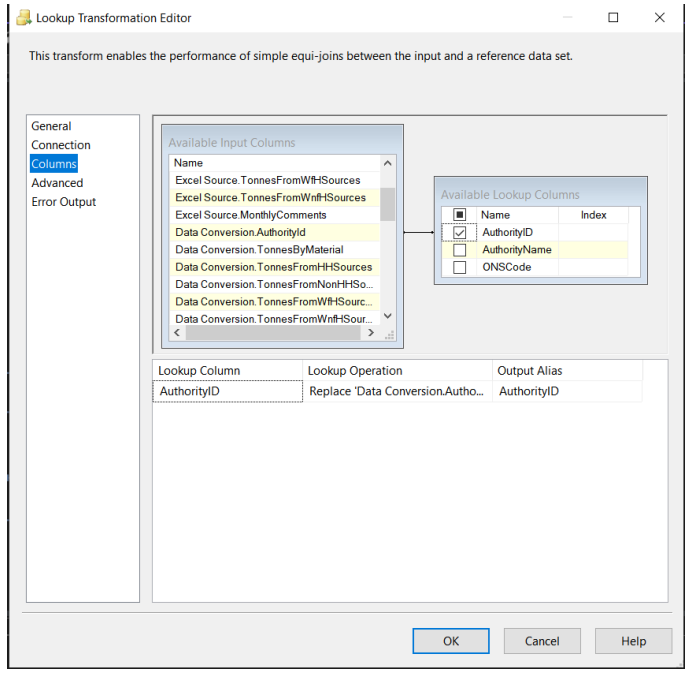
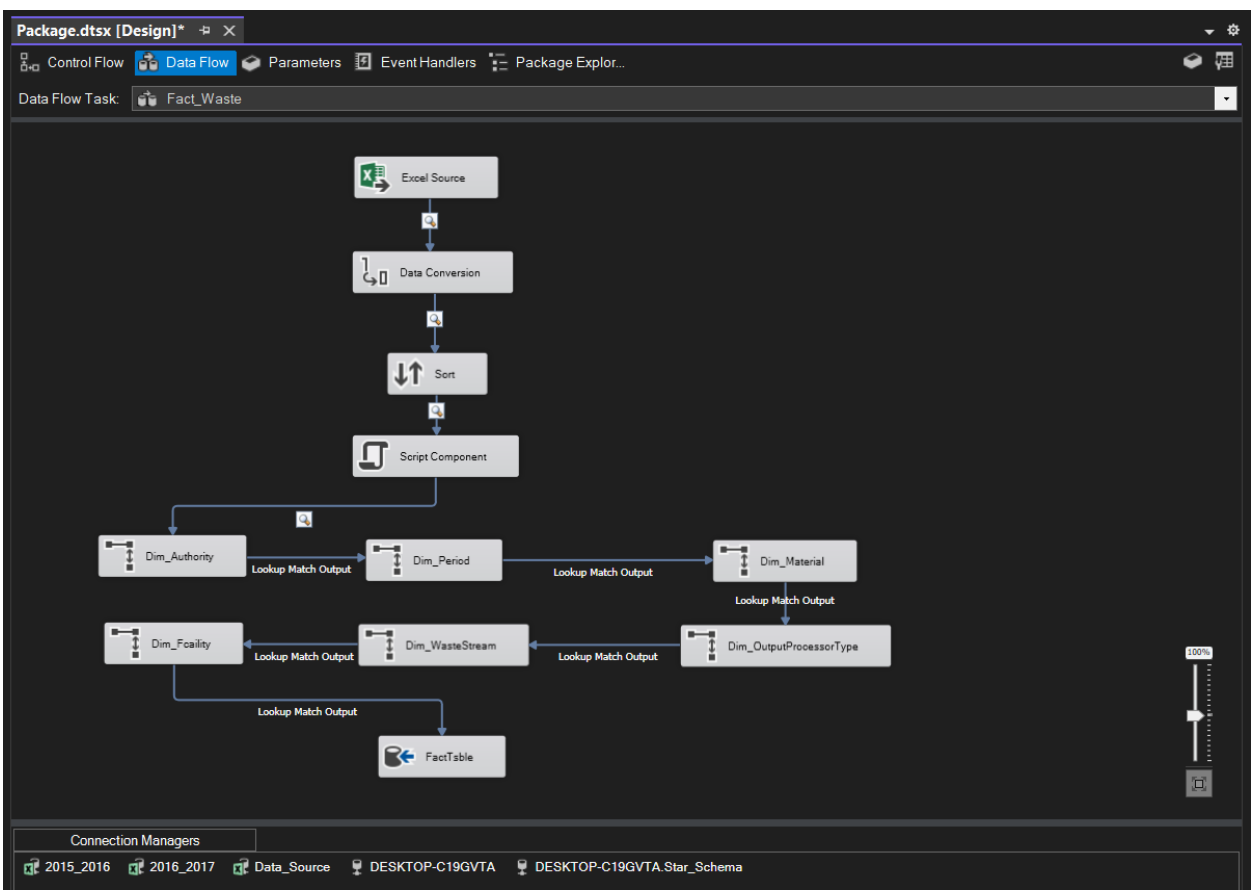
Attach...
Data Sources
SSIS Toolbox
This tool window can only be used by a SQL Server Integration Services package document.
main.cs
SC_51c3b33c42504f8ca1421b5d07a55c7a
ScriptMain
PreExecute()
* Add your code here
*/
}
/// <summary>
/// This method is called after all the rows have passed through this component.
///
/// You can delete this method if you don't need to do anything here.
/// </summary>
0 references
public override void PostExecute()
{
base.PostExecute();
/*
* Add your code here
*/
}
/// <summary>
/// This method is called once for every row that passes through the component from Input0.
///
/// Example of reading a value from a column in the the row:
/// string zipCode = Row.ZipCode
///
/// Example of writing a value to a column in the row:
/// Row.ZipCode = zipCode
/// </summary>
/// <param name="Row">The row that is currently passing through the component</param>
///
int i = 0;
2 references
public override void Input0_ProcessInputRow(Input0Buffer Row)
{
Row.FactId = i + 1;
i = i + 1;
}
}

4. Utilized **Lookup Transformations** to establish relationships with dimension tables.





5. Populated the fact table with the cleaned data into the SSMS database.



B. SQL Query Tasks

With the database populated, the following SQL queries were executed to derive insights:

1. **Top 5 Waste Materials**:
   Showed the top 5 “Dry Recyclate” waste materials processed by each authority, grouped by quarter, with total tonnage for each period (e.g., January-March, April-June).

```sql
/*4a: Show the top 5 “Dry Recyclate” waste materials processed by each authority
and the total weight in TotalTonnes by quarters e.g. April-June, JulySeptember, October-Decei
top 5 materials may vary)*/


WITH RankedMaterials AS (
    SELECT
        A.AuthorityName,
        P.Quarter,
        P.Year,
        M.MaterialGroup,
        SUM(F.TonnesByMaterial) AS TotalTonnes,
        RANK() OVER (
            PARTITION BY A.AuthorityName, P.Quarter
            ORDER BY SUM(F.TonnesByMaterial) DESC
        ) AS Rank
    FROM
        Fact_WasteManagement F
    INNER JOIN Dim_Authority A ON F.AuthorityID = A.AuthorityID
    INNER JOIN Dim_Period P ON F.PeriodID = P.PeriodID
    INNER JOIN Dim_Material M ON F.MaterialID = M.MaterialID
    INNER JOIN Dim_OutputProcessType O ON F.OutputProcessTypeID=O.OutputProcessTypeID
    WHERE
        O.OutputProcessType = 'Dry Recyclate'
    GROUP BY
        A.AuthorityName, P.Quarter, Year,M.MaterialGroup
)
SELECT
    AuthorityName,
    Quarter,
    Year,
    MaterialGroup,
    TotalTonnes
FROM
    RankedMaterials
WHERE
    Rank <= 5
ORDER BY
    AuthorityName, Quarter, Year,Rank;
```

%

Results | Messages

AuthorityName	Quarter	Year	MaterialGroup	TotalTonnes
Adur District Council	Q1	2016	Paper & Card	2137.77
Adur District Council	Q1	2016	Glass	723.02
Adur District Council	Q1	2016	Plastic	231.16
Adur District Council	Q1	2017	Paper & Card	1927.19
Adur District Council	Q1	2017	Glass	767.84
Adur District Council	Q2	2015	Paper & Card	1923.93
Adur District Council	Q2	2015	Glass	773.12

2. **Top 10 Authorities**:
   Ranked the top 10 authorities by total tonnage processed for each six-month period, displaying changes in ranking from the previous period.

```sql
/*4b: ank the top 10 authorities and the TotalTonnes of waste processed by each
of these for each six-month period. For the second period onward also show
the number of places each Authority has moved (up/down the ranking) since
the previous period.*/

WITH RankedAuthorities AS (
    SELECT
        DP.SixMonthPeriod,
        DA.AuthorityName,
        SUM(FWM.TotalTonnes) AS TotalTonnes,
        RANK() OVER (PARTITION BY DP.SixMonthPeriod ORDER BY SUM(FWM.TotalTonnes) DESC) AS CurrentRank
    FROM
        Fact_WasteManagement FWM
        JOIN Dim_Authority DA ON FWM.AuthorityID = DA.AuthorityID
        JOIN Dim_Period DP ON FWM.PeriodID = DP.PeriodID
    GROUP BY
        DP.SixMonthPeriod,
        DA.AuthorityName
),
RankMovement AS (
    SELECT
        RA1.SixMonthPeriod,
        RA1.AuthorityName,
        RA1.TotalTonnes,
        RA1.CurrentRank,
        RA2.CurrentRank AS PreviousRank,
        COALESCE(RA2.CurrentRank - RA1.CurrentRank, 0) AS RankMovement
    FROM
        RankedAuthorities RA1
        LEFT JOIN RankedAuthorities RA2
            ON RA1.AuthorityName = RA2.AuthorityName
            AND RA2.SixMonthPeriod = (
                SELECT MAX(DP2.SixMonthPeriod)
                FROM Dim_Period DP1
                JOIN Dim_Period DP2 ON DP1.SixMonthPeriod > DP2.SixMonthPeriod
                WHERE DP1.SixMonthPeriod = RA1.SixMonthPeriod
            )
)
SELECT
    SixMonthPeriod,
    AuthorityName,
    TotalTonnes,
    CurrentRank,
    PreviousRank,
    RankMovement
FROM
    RankMovement
WHERE
    CurrentRank <= 10
ORDER BY
    SixMonthPeriod, CurrentRank;
```

56 %

Results | Messages

	SixMonthPeriod	AuthorityName	TotalTonnes	CurrentRank	PreviousRank	RankMovement
1	H1	Greater Manchester WDA (MBC)	2006398.416	1	NULL	0
2	H1	Kent County Council	1341992.82	2	NULL	0
3	H1	Hampshire County Council	1323596.95	3	NULL	0
4	H1	North London Waste Authority	1274778.62	4	NULL	0
5	H1	Birmingham City Council	1176850.2	5	NULL	0

Query executed successfully.

3. **Facility Type and WasteStream Type**:
   For each facility type within each authority, the total tonnage from households (HH) was shown, grouped by waste stream type. Then the output was exported to a JSON file for reporting.

```sql
/*4c: For each FacilityType in each authority, show the WasteStreamType,
TotalTonnesFromHH (households) reported for the two-year period order the
output in descending order of TotalTonnesFromHH. Export the query output as
a JSON file.*/

SELECT
    A.AuthorityName,
    F.FacilityType,
    W.WasteStreamType,
    SUM(FM.TonnesFromHHSources) AS TotalTonnesFromHH
FROM
    Fact_WasteManagement FM
INNER JOIN Dim_Authority A ON FM.AuthorityID = A.AuthorityID
INNER JOIN Dim_Facility F ON FM.FacilityTypeID = F.FacilityTypeID
INNER JOIN Dim_WasteStream W ON FM.WasteStreamID = W.WasteStreamID
INNER JOIN Dim_Period P ON FM.PeriodID = P.PeriodID
WHERE
    p.Year >= (SELECT MAX(Year) - 1 FROM Dim_Period)    -- Last Two-year period
GROUP BY
    A.AuthorityName,
    F.FacilityType,
    W.WasteStreamType
ORDER BY
    TotalTonnesFromHH DESC
FOR JSON PATH, ROOT('FacilityData');
```

93 %

Results | Messages

	JSON_F52E2B61-18A1-11d1-B105-00805F49916B
1	{"FacilityData":[{"AuthorityName":"Lancashire Count...

4. **Dynamic Query via Stored Procedure**:
   A stored procedure was created to accept user input and return results based on a time-related hierarchy, demonstrating advanced SQL query writing.

```sql
/*Create a fourth dynamic query as a stored procedure that accepts input
parameters, utilizing a time related hierarchy of your own design that
demonstrates the full complexity of your understanding in terms of analytic
query writing*/

CREATE PROCEDURE GetsWasteDataByTimeHierarchy
    @StartYear INT = NULL,              -- Optional parameter for the start year
    @EndYear INT = NULL,                -- Optional parameter for the end year
    @Quarter NVARCHAR(2) = NULL,        -- Optional parameter for the quarter (Q1, Q2, Q3, Q4)
    @SixMonthPeriod NVARCHAR(10) = NULL -- Optional parameter for the six-month period (H1, H2)
AS
BEGIN
    DECLARE @SQLQuery NVARCHAR(MAX);

    -- Base query to get waste data by time hierarchy
    SET @SQLQuery = '
    SELECT
        a.AuthorityName,
        f.FacilityType,
        ws.WasteStreamType,
        SUM(fwm.TotalTonnes) AS TotalTonnes,
        SUM(fwm.TonnesByMaterial) AS TotalTonnesByMaterial,
        p.Year,
        p.Quarter,
        p.SixMonthPeriod
    FROM
        Fact_WasteManagement fwm
    INNER JOIN
        Dim_Authority a ON fwm.AuthorityID = a.AuthorityID
    INNER JOIN
        Dim_Facility f ON fwm.FacilityTypeID = f.FacilityTypeID
    INNER JOIN
        Dim_WasteStream ws ON fwm.WasteStreamID = ws.WasteStreamID
    INNER JOIN
        Dim_Period p ON fwm.PeriodID = p.PeriodID
    WHERE 1=1';  -- Placeholder to append conditions dynamically

    -- Dynamically add conditions based on provided input parameters
    IF @StartYear IS NOT NULL AND @EndYear IS NOT NULL
    BEGIN
        SET @SQLQuery = @SQLQuery + ' AND p.Year BETWEEN ' + CAST(@StartYear AS NVARCHAR) + ' AND ' + CAST(@EndYear AS NVARCHAR);
    END
    ELSE IF @StartYear IS NOT NULL
    BEGIN
        SET @SQLQuery = @SQLQuery + ' AND p.Year = ' + CAST(@StartYear AS NVARCHAR);
    END
    ELSE IF @EndYear IS NOT NULL
    BEGIN
        SET @SQLQuery = @SQLQuery + ' AND p.Year = ' + CAST(@EndYear AS NVARCHAR);
    END

    IF @Quarter IS NOT NULL
    BEGIN
        SET @SQLQuery = @SQLQuery + ' AND p.Quarter = ''' + @Quarter + '''';
    END

    IF @SixMonthPeriod IS NOT NULL
    BEGIN
        SET @SQLQuery = @SQLQuery + ' AND p.SixMonthPeriod = ''' + @SixMonthPeriod + '''';
    END

    -- Grouping and ordering the results
    SET @SQLQuery = @SQLQuery + '
    GROUP BY
        a.AuthorityName,
        f.FacilityType,
        ws.WasteStreamType,
        p.Year,
        p.Quarter,
        p.SixMonthPeriod
    ORDER BY
        TotalTonnes DESC';

    -- Execute the dynamically constructed SQL query
    EXEC sp_executesql @SQLQuery;
END;
```

43 %

Results | Messages

	AuthorityName	FacilityType	WasteStreamType	Tot
1	North London Waste Authority	Incineration with energy recovery	Residual waste	14

Query executed successfully.

3. Visualization Using Power BI

An interactive Power BI dashboard was created to visualize the results of the SQL queries. The dashboard provides an in-depth analysis of waste material recycling data for the selected quarter (**Q1, Q2, Q3, Q4**) and half-year period (**H1** ,**H2**), with the option to filter data using a slider for years (2015-2017). Here's a breakdown of each section:

- **Trends over Time**: Seasonal and annual trends in tonnage processed were clearly identified.
- **Top Authorities**: A leaderboard-style visualization highlighted the top 10 authorities by waste processed for each period, with tooltips displaying rank changes.

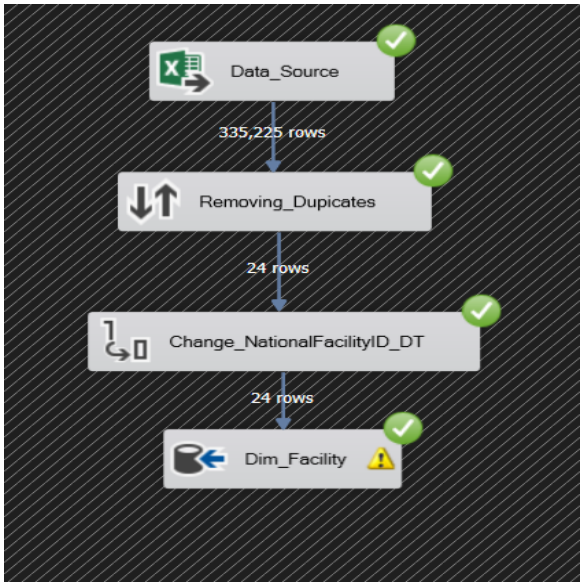
- **Material Analysis**: A drill-down feature allowed users to explore top waste materials by authority and time period.
- **Facility Analysis**: Bar graphs displayed the distribution of waste stream types by facility type.

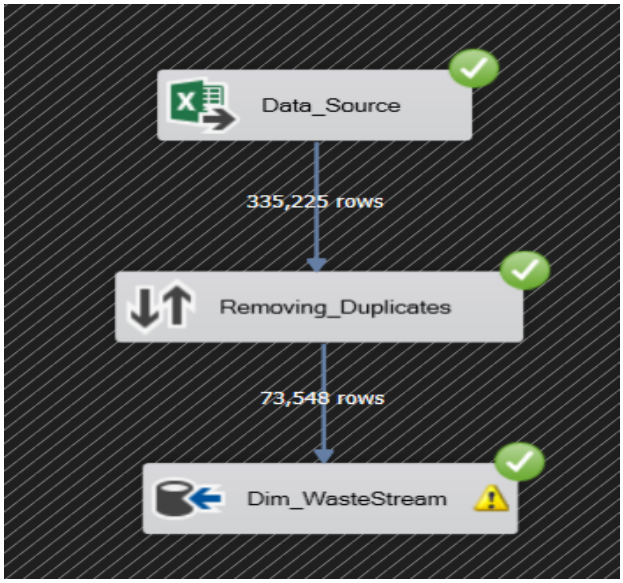
VI. Results and Analysis:

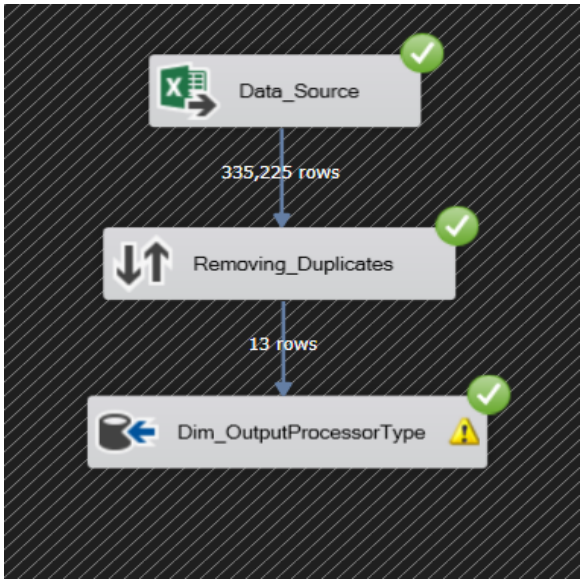
The results of the project highlight key insights and trends derived from the cleaned and transformed data stored in the data warehouse. Using the interactive Power BI dashboard, as well as the SQL query outputs, several patterns and actionable insights were identified.

A. ETL

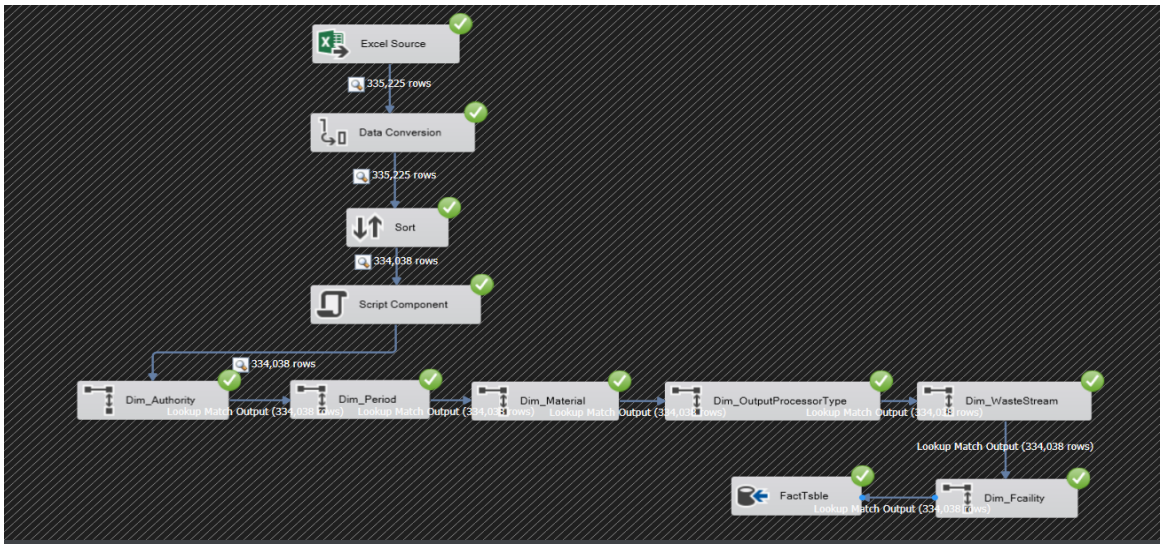
All the dimensions and fact table successfully populated accurately mapped to their respective column in SSMS database. Dimension tables such as Dim_Authority, Dim_Period, and Dim_Material were successfully populated after data cleaning, duplication removal, and transformations. For example, the







Dim_Period table was enriched with derived fields such as Quarter, SixMonthPeriod, and Year to facilitate time-based analysis. The fact table Fact_WasteManagement was populated with optimized data types for memory efficiency, and relationships between the fact and dimension tables were established through lookups.



Screenshots of the running data flows for each dimension and fact table in SSIS validated the successful execution of the ETL processes, providing a clear picture of the project's technical implementation.

B. Query Results and Insights

a. Top 5 “Dry Recyclate” Waste Materials by Authority and Quarter

Each authority processed different types of "Dry Recyclate" waste materials as the top 5 in each quarter, resulting a seasonal variations in recycling trends. Also materials such as **paper and cards, plastics, and glass** were consistently among the top items for most authorities.

```sql
/*4a: Show the top 5 "Dry Recyclate" waste materials processed by each authority
and the total weight in TotalTonnes by quarters e.g. April-June, JulySeptember, October-Dece
top 5 materials may vary)*/


WITH RankedMaterials AS (
    SELECT
        A.AuthorityName,
        P.Quarter,
        P.Year,
        M.MaterialGroup,
        SUM(F.TonnesByMaterial) AS TotalTonnes,
        RANK() OVER (
            PARTITION BY A.AuthorityName, P.Quarter
            ORDER BY SUM(F.TonnesByMaterial) DESC
        ) AS Rank
    FROM
        Fact_WasteManagement F
    INNER JOIN Dim_Authority A ON F.AuthorityID = A.AuthorityID
    INNER JOIN Dim_Period P ON F.PeriodID = P.PeriodID
    INNER JOIN Dim_Material M ON F.MaterialID = M.MaterialID
    INNER JOIN Dim_OutputProcessType O ON F.OutputProcessTypeID=O.OutputProcessTypeID
    WHERE
        O.OutputProcessType = 'Dry Recyclate'
    GROUP BY
        A.AuthorityName, P.Quarter, Year,M.MaterialGroup
)
SELECT
    AuthorityName,
    Quarter,
    Year,
    MaterialGroup,
    TotalTonnes
FROM
    RankedMaterials
WHERE
    Rank <= 5
ORDER BY
    AuthorityName, Quarter, Year,Rank;
```

Results | Messages

AuthorityName	Quarter	Year	MaterialGroup	TotalTonnes
Adur District Council	Q1	2016	Paper & Card	2137.77
Adur District Council	Q1	2016	Glass	723.02
Adur District Council	Q1	2016	Plastic	231.16
Adur District Council	Q1	2017	Paper & Card	1927.19
Adur District Council	Q1	2017	Glass	767.84
Adur District Council	Q2	2015	Paper & Card	1923.93
Adur District Council	Q2	2015	Glass	773.12

Results | Messages

	AuthorityName	Quarter	Year	MaterialGroup	TotalTonnes
1	Adur District Council	Q1	2016	Paper & Card	2137.77
2	Adur District Council	Q1	2016	Glass	723.02
3	Adur District Council	Q1	2016	Plastic	231.16
4	Adur District Council	Q1	2017	Paper & Card	1927.19
5	Adur District Council	Q1	2017	Glass	767.84
6	Adur District Council	Q2	2015	Paper & Card	1923.93
7	Adur District Council	Q2	2015	Glass	773.12
8	Adur District Council	Q2	2016	Paper & Card	1425.46
9	Adur District Council	Q2	2016	Glass	798.14
10	Adur District Council	Q2	2016	Plastic	216.7
11	Adur District Council	Q3	2015	Paper & Card	1958.56
12	Adur District Council	Q3	2015	Glass	736.1
13	Adur District Council	Q3	2016	Paper & Card	1922.8
14	Adur District Council	Q3	2016	Glass	846.04
15	Adur District Council	Q3	2016	Plastic	264.59
16	Adur District Council	Q4	2015	Paper & Card	2139.29
17	Adur District Council	Q4	2015	Glass	759.88
18	Adur District Council	Q4	2016	Paper & Card	2084.94
19	Adur District Council	Q4	2016	Glass	765.22
20	Adur District Council	Q4	2016	Plastic	202.26
21	Allerdale Borough ...	Q1	2016	Paper & Card	1104.58
22	Allerdale Borough ...	Q1	2016	Glass	1072.72
23	Allerdale Borough ...	Q1	2016	Plastic	468.16
24	Allerdale Borough ...	Q1	2017	Glass	1188.22
25	Allerdale Borough ...	Q1	2017	Paper & Card	1067.64
26	Allerdale Borough ...	Q2	2015	Paper & Card	1150.1
27	Allerdale Borough ...	Q2	2015	Glass	1079.26

Query executed successfully.

b. **Top 10 Authorities by Total Tonnes Processed for Each Six-Month Period**
Authorities with the highest processing capacities remained consistent across periods. However, some mid-tier authorities showed significant shifts in ranking. The top-performing authorities, such as **Greater Manchester WDA (MBC)** maintained stable tonnage outputs across all six-month periods.

	SixMonthPeriod	AuthorityName	TotalTonnes	CurrentRank	PreviousRank	RankMovement
1	H1	Greater Manchester WDA (MBC)	2006398.416	1	NULL	0
2	H1	Kent County Council	1341992.82	2	NULL	0
3	H1	Hampshire County Council	1323596.95	3	NULL	0
4	H1	North London Waste Authority	1274778.62	4	NULL	0
5	H1	Birmingham City Council	1176850.2	5	NULL	0
6	H1	Merseyside WDA (MBC)	1069517.69	6	NULL	0
7	H1	Essex County Council	989609.771	7	NULL	0
8	H1	Leeds City Council MBC	989088.669000001	8	NULL	0
9	H1	West London Waste Authority	967538.14	9	NULL	0
10	H1	Lancashire County Council	948334.423	10	NULL	0
11	H2	Greater Manchester WDA (MBC)	1845671.298	1	1	0
12	H2	North London Waste Authority	1328713.49	2	4	2
13	H2	Hampshire County Council	1273114.03	3	3	0
14	H2	Kent County Council	1266931.25	4	2	-2
15	H2	Birmingham City Council	1160281.05	5	5	0
16	H2	Merseyside WDA (MBC)	1059120.052	6	6	0
17	H2	West London Waste Authority	975945.6	7	9	2
18	H2	Leeds City Council MBC	960818.993	8	8	0
19	H2	Lancashire County Council	928855.584	9	10	1
20	H2	Western Riverside Waste Auth...	919904.93	10	12	2

Query executed successfully.

c. **Waste Stream Type by Facility Type and Authority**

Each facility type managed distinct waste stream types, with household (HH) waste accounting for the majority of the total tonnage.

4c_query.json - Notepad

File Edit Format View Help


```
{"FacilityData":[{"AuthorityName":"Lancashire County Council","FacilityType":"Fina
shire County Council","FacilityType":"Reprocessor - recycling (qu19)","WasteStream
":"Reprocessor - recycling (qu19)","WasteStreamType":"Comingled recyclate","TotalT
lTonnesFromHH":8.757060700000001e+005},{"AuthorityName":"South Gloucestershire Cou
"Ealing LB","FacilityType":"Final Destination","WasteStreamType":"Comingled recycl
ation","WasteStreamType":"Comingled recyclate","TotalTonnesFromHH":6.9381055999999
"WasteStreamType":"Residual waste","TotalTonnesFromHH":6.431849200000000e+005},{"A
horityName":"Southend-on-Sea Borough Council","FacilityType":"Final Destination","
ste Authority","FacilityType":"Final Destination","WasteStreamType":"Residual wast
mType":"Residual waste","TotalTonnesFromHH":5.151044400000001e+005},{"AuthorityNam
nnesFromHH":4.674030000000000e+005},{"AuthorityName":"Sunderland City Council","Fa
,"FacilityType":"Reprocessor - recycling (qu19)","WasteStreamType":"Comingled recy
":"South Oxfordshire District Council","FacilityType":"Reprocessor - recycling (qu
thorityName":"Uttlesford District Council","FacilityType":"Reprocessor - recycling
"Reprocessor - recycling (qu19)","WasteStreamType":"Comingled recyclate","TotalTon
"WasteStreamType":"Comingled recyclate","TotalTonnesFromHH":3.863745800000004e+005
horityName":"Vale of White Horse District Council","FacilityType":"Final Destinati
l","FacilityType":"Incineration with energy recovery","WasteStreamType":"Residual
ouncil","FacilityType":"Final Destination","WasteStreamType":"Comingled recyclate"
ing (qu19)","WasteStreamType":"Comingled recyclate","TotalTonnesFromHH":3.25755050
gled recyclate","TotalTonnesFromHH":3.184500600000001e+005},{"AuthorityName":"Adur
00001e+005},{"AuthorityName":"Leeds City Council MBC","FacilityType":"Incineration
"FacilityType":"Incineration with energy recovery","WasteStreamType":"Residual was
sor - recycling (qu19)","WasteStreamType":"Comingled recyclate","TotalTonnesFromHH
"Exeter City Council","FacilityType":"Final Destination","WasteStreamType":"Coming
ssor - recycling (qu19)","WasteStreamType":"Comingled recyclate","TotalTonnesFromH
lTonnesFromHH":2.563954000000000e+005},{"AuthorityName":"Walsall MBC","FacilityTyp
d","FacilityType":"Final Destination","WasteStreamType":"Comingled recyclate","Tot
alTonnesFromHH":2.404025500000000e+005},{"AuthorityName":"Eastleigh Borough Counci
"AuthorityName":"South Norfolk Council","FacilityType":"Final Destination","WasteS
thorityName":"Plymouth City Council","FacilityType":"Final Destination","WasteStre
sor - recycling (qu19)","WasteStreamType":"Residual waste","TotalTonnesFromHH":2.1
:"Comingled recyclate","TotalTonnesFromHH":2.101048799999997e+005},{"AuthorityName
},{"AuthorityName":"Leicestershire County Council","FacilityType":"Incineration wi
ncolnshire County Council","FacilityType":"Hazardous landfill","WasteStreamType":"
C","FacilityType":"Reprocessor - recycling (qu19)","WasteStreamType":"Residual was
ype":"Final Destination","WasteStreamType":"Residual waste","TotalTonnesFromHH":1.
very","WasteStreamType":"Residual waste","TotalTonnesFromHH":1.912757400000000e+00
```


d. **Dynamic Query via Stored Procedure**:
A stored procedure was created to accept user input and return results based on a time-related hierarchy, demonstrating advanced SQL query writing.

43 %

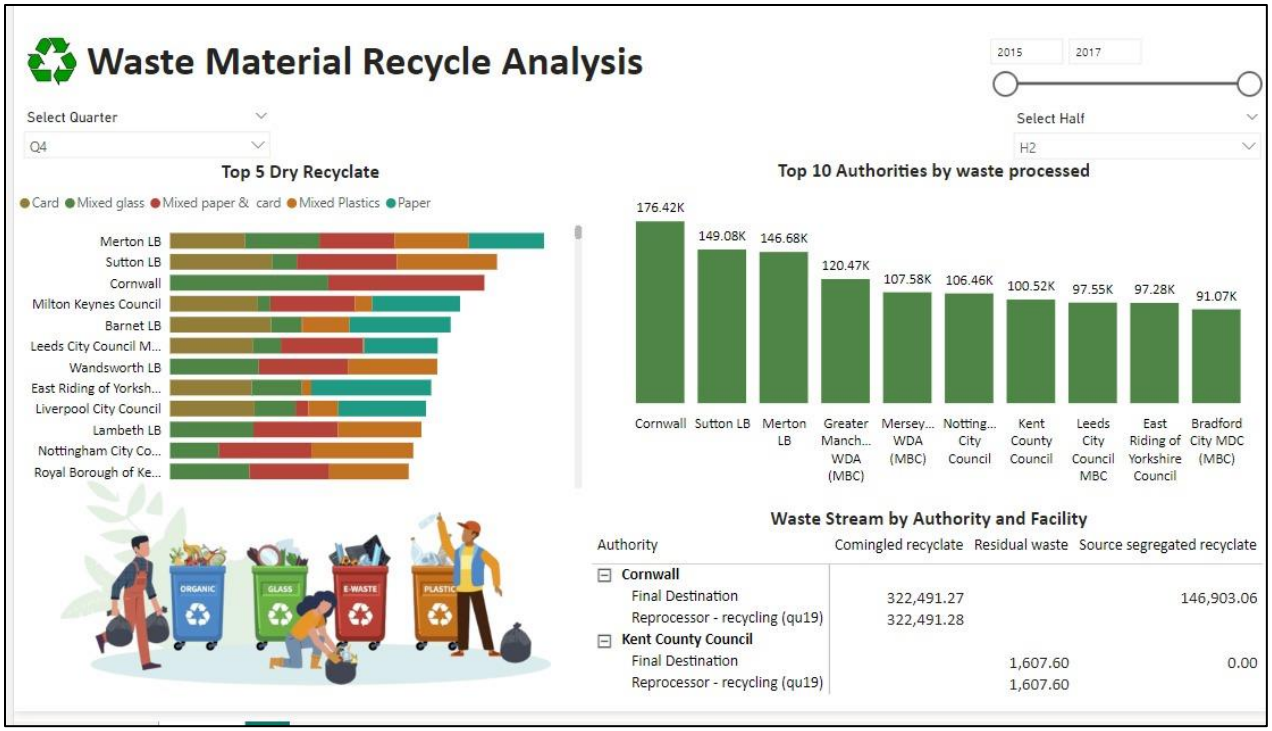
Results | Messages

	AuthorityName	FacilityType	WasteStreamType	TotalTonnes	TotalTonnesByMaterial	Year	Quarter	SixMo
1	North London Waste Authority	Incineration with energy recovery	Residual waste	141872.32	0	2016	Q3	H2
2	North London Waste Authority	Incineration with energy recovery	Residual waste	115639.68	0	2016	Q4	H2
3	Greater Manchester WDA (MBC)	Incineration with energy recovery	Residual waste	112901.71	0	2016	Q3	H2
4	North London Waste Authority	NULL	Residual waste	111165.41	0	2016	Q3	H2
5	Greater Manchester WDA (MBC)	Incineration with energy recovery	Residual waste	104289.1	0	2016	Q4	H2
6	West London Waste Authority	Incineration with energy recovery	Residual waste	96008.6	0	2016	Q4	H2
7	North London Waste Authority	NULL	Residual waste	95809.07	0	2016	Q4	H2
8	Kent County Council	Incineration with energy recovery	Residual waste	90511.34	0	2016	Q3	H2
9	Greater Manchester WDA (MBC)	Mechanical Biological treatment	Residual waste	90222.82	0	2016	Q3	H2
10	West London Waste Authority	NULL	Residual waste	87301.45	0	2016	Q4	H2
11	Hampshire County Council	Incineration with energy recovery	Residual waste	87181.85	0	2016	Q3	H2
12	West London Waste Authority	Incineration with energy recovery	Residual waste	86751.23	0	2016	Q3	H2
13	Birmingham City Council	Incineration with energy recovery	Residual waste	86260.37	0	2016	Q4	H2
14	Greater Manchester WDA (MBC)	NULL	Residual waste	86048.64	0	2016	Q3	H2
15	Hampshire County Council	Final Destination	Source segregated recyclate	85319.52	23210.16	2016	Q3	H2
16	Kent County Council	Incineration with energy recovery	Residual waste	84010.44	0	2016	Q4	H2
17	Lancashire County Council	Non-hazardous landfill	Residual waste	82874.128	0	2016	Q3	H2
18	Hampshire County Council	Incineration with energy recovery	Residual waste	81335.13	0	2016	Q4	H2
19	West London Waste Authority	NULL	Residual waste	80358.71	0	2016	Q3	H2
20	Greater Manchester WDA (MBC)	NULL	Residual waste	79709.8	0	2016	Q4	H2
21	Hampshire County Council	Reprocessor - recycling (qu19)	Source segregated recyclate	79696	21743.8	2016	Q3	H2
22	Western Riverside Waste Autho...	Incineration with energy recovery	Residual waste	78896.43	0	2016	Q3	H2
23	Birmingham City Council	Incineration with energy recovery	Residual waste	77586.95	0	2016	Q3	H2
24	Western Riverside Waste Autho...	Incineration with energy recovery	Residual waste	76704.95	0	2016	Q4	H2
25	Greater Manchester WDA (MBC)	Mechanical Biological treatment	Residual waste	74453.56	0	2016	Q4	H2
26	Kent County Council	NULL	Residual waste	70929.42	0	2016	Q3	H2
27	East London Waste Authority	NULL	Residual waste	70350	0	2016	Q3	H2
28	Hampshire County Council	NULL	Residual waste	70268.28	0	2016	Q3	H2
29	East London Waste Authority	Mechanical Biological treatment	Residual waste	70246	0	2016	Q3	H2

Query executed successfully.

- **2. Power BI Dashboard Analysis**

The data gathered through SQL queries was visualized in an interactive Power BI dashboard, combining all four analysis tasks into a single view. This dashboard enables stakeholders to explore trends, compare authority performances, and make data-driven decisions.



VII. Conclusion

This project successfully demonstrates the process of designing and implementing a data warehouse solution using Kimball's Dimensional Modelling Technique, SSIS for ETL, SSMS for querying, and Power BI for visualization, starting from handling raw data to creating a fully functional database with insights ready for analysis. The project began with resolving issues in the source data, such as null values, column mismatches, and duplicates, datatype conversions, and dynamic indexing, enhancing the overall reliability and scalability of the system. Using SSIS, developed data flows to transform and load data into well-structured fact and dimension tables following the star schema model. By writing SQL queries, extracted meaningful insights, such as waste management trends and authority rankings, and showcased them in an interactive Power BI dashboard. This not only highlights the potential of data warehousing for analysis but also shows how such systems can be used to make informed decisions.

Overall, this project gave the hands-on experience in data integration, ETL processes, database management, and data visualization. It demonstrates the importance of a well-designed data warehouse in turning raw data into valuable insights while laying the groundwork for further enhancements in the future.

VIII. Future Work

While the project successfully met its objectives, but in the future, this project can be improved in several ways. One important step is to deploy the data warehousing solution using SSIS. This would allow automated scheduling and execution of ETL processes, making data updates more efficient and less time-consuming and also, we can automate ETL workflows using Azure Data Factory. The project could also benefit from integrating machine learning models to predict trends based on the data stored in the warehouse. Another improvement would be to enhance the Power BI dashboard by adding real-time

analytics and more interactive features such as Adding geospatial analytics to map waste processing facilities and visualize authority-wise waste distribution on an interactive map in Power BI. Also, developing KPI-driven dashboards to monitor real-time waste management would be great and making the dashboard even more user-friendly and insightful. Develop a user-friendly web application for broader access to the data warehouse insights by non-technical stakeholders would be an another approach. Additionally, incorporating other relevant datasets, such as environmental or socio-economic data, could provide more comprehensive insights and expand the analytical capabilities of the project. These enhancements would not only make the project more robust and scalable but also ensure its practical application in real-world scenarios.